Q1 Describe the pyrolysis process & incineration process.

Ans:

- Pyrolysis is a thermal decomposition process in which organic solid waste is heated at high temperatures in absence or limited supply of oxygen.
- This technique is used for volume reduction and energy recovery.
- The waste materials are decomposed by heat without combustion due to absence of oxygen.
- Complex organic substances break into simpler gases & liquids.

Steps:

1. Segregation: Removal of metals & inert materials
2. Shredding & drying: Waste is reduced in size and moisture content.
3. Heating: Waste is heated to 300-800°C in closed reactor without oxygen.
4. Thermal decomposition: Organic matter decomposes into char, oil and gases.
5. Collection of products: Gas and oil are collected for energy recovery.

- Products of pyrolysis: char, tar, hydrocarbons, methane, CO, hydrogen.
- This method is used in treatment of plastic waste, biomass & organic waste processing, energy recovery from solid waste.

- Incineration is a controlled combustion process in which solid waste is burned in presence of excess oxygen at high temperature.
- This is done to reduce waste volume & destroy pathogens.

- Combustible waste reacts with oxygen producing heat, ash, ~~fuel~~ gases. flue gases
- The heat generated can be used for steam and electricity generation.

Steps:

1. Collection of waste
2. Waste sent to furnace
3. Combustion at 850-1100°C with excess oxygen.
4. Organic matter burns & produces heat & gases.
5. Residue ash is collected and disposed.
6. Flue gases are treated before release.

- Products of incineration: Ash, flue gases, heat energy.
- Used in municipal solid waste treatment, biomedical waste disposal, hazardous waste treatment and energy recovery plants.

Q.2 Explain the general effects of solid waste on health and environment.

Ans:

- Solid waste generated from domestic, industrial and commercial activities has significant impact on health and environment.
- Improper handling, storage and disposal of solid waste leads to pollution, spread of diseases and ecological balance.

Effects on health:

1. Spread of diseases: Uncollected waste attracts flies, mosquitos, rats leading to diseases like malaria, dengue, cholera etc.

2. Air pollution and respiratory problems: Burning of solid waste releases gases like CO, SO_2, particulate matter that cause asthma, bronchitis, lung infections etc.

3. Waterborne diseases: Leachate from waste dumps contaminates surface & groundwater. This leads to diarrhoea, dysentery, hepatitis etc.

4. Occupational hazards: Waste handling workers are exposed to sharp objects, toxic chemicals, skin diseases etc.

5. Hazardous waste can also lead to cancer, nervous system damage and kidney problems.

Effects on environment:

1. Open dumping of waste degrades land quality and reduces soil fertility.
2. Leachate from landfills contaminates rivers, lakes and groundwater. This affects aquatic life.
3. Decomposition of organic waste produces gases like methane and CO_2.
4. Methane gas released from landfills contributes to global warming and climate change.
5. Animals ingest plastic and die.

Q.3 Explain in details the ultimate and proximate analysis of solid waste.

Ans: Proximate Analysis:

- Provides physical composition of solid waste. It provides info about moisture content, volatile matter, fixed carbon & ash content.

Components:

1. Moisture content: Amount of water in waste.
2. Volatile matter: Volatile matter consists of organic substances that vaporize when heated in absence of O_2.
3. Ash content: Non-combustible residue left after burning of waste.
4. Fixed Carbon: Solid combustible residue left after removing volatile matter.

- Proximate Analysis helps in design of incinerators, determines fuel characteristics etc.

Ultimate Analysis of Solid Waste:

- Ultimate analysis determines the chemical composition of solid waste. It measure percentage of elements in the waste.

Components:

1. Carbon:
2. Hydrogen
3. Oxygen
4. Nitrogen
5. Sulphur
6. Ash.

- Ultimate Analysis determines chemical composition of waste, helps in energy recovery calculations, used for combustion & incineration design.

Q.4 Discuss about municipal, industrial & hazardous waste.

Ans:

- Solid waste is generated from various sources.
- Based on the source and nature, solid waste is classified into municipal waste, industrial waste and hazardous waste.

Municipal Solid Waste:

- Waste generated from residential, commercial, institutional and public places in urban areas.
- Sources: households, market places, schools, hospitals, streets, parks
- Components: food waste, paper and cardboard, plastics, glass, metals, ash & dust.
- These wastes are mostly biodegradable and recyclable. It contains organic & inorganic matter.

Industrial Waste:

- Waste generated from manufacturing and industrial processes.
- Sources: Chemical industries, textile industries, oil refineries, food processing units.
- Components: scrap, slag, effluents, emissions.
- These wastes contain toxic chemicals, and are generated in large quantities.

Hazardous wastes:

- Waste that poses danger to human health and the environment due to its toxic, corrosive, reactive and flammable nature.
- Sources: Industries, hospitals, Laboratories, paint & battery industries.
- These wastes are toxic, corrosive, reactive, flammable, radioactive.

Q.5 Explain in detail the concept of on-site handling of solid waste.

Ans:

- On-site handling of solid waste refers to all activities associated with handling, segregation, storage and processing of solid waste at the point where it is generated before collection.

- Its objectives include:
  → Reduce waste generation at source.
  → Facilitate segregation & recycling.
  → Prevent health hazards.
  → Improve collection efficiency

- Elements of On-Site Handling:
  1. Waste Segregation at source: separation of waste into diff. categories like biodegradable, non-biodegradable, hazardous.
  2. Waste Storage at source: temporary holding in household bins, commercial storage bins etc.
  3. Reuse of materials: Materials like glass bottles, plastic can be reused without processing.
  4. Recycling at source: Paper, plastic, metal, glass can be recycled

Q.6 Describe factors affecting selection of storage containers

Ans:

- Storage of solid waste is an important step in solid waste management.
- Storage containers are used to temporarily hold waste at the source before collection.

Factors :

1. Quantity of waste generated : Large & small.

2. Type of waste : Biodegradable, hazardous, dry, wet.

3. Frequency of collection : Daily, weekly, irregular.

4. Location of storage : Residential, commercial, public

5. Material of container : Plastic, metal, fiberglass.

6. Handling & Transportation.

7. Protection against animals & insects.

8. Cost and maintenance :

Q.7 Explain mechanical volume reduction.

Ans :

- Mechanical volume reduction is the process of reducing volume of solid waste by using mechanical methods without changing its chemical composition.

Objectives of mechanical volume reduction:

1. Reduce volume of solid waste
2. Facilitate easy handling & transport
3. Increase landfill life.
4. Improve efficiency of treatment processes.
5. Reduce transportation cost.

Methods:

1. Compaction
2. Baling
3. Shredding
4. Grinding & Pulverization